MSG No. 164.87 $P\bar{3}'m1$ [Type III, trigonal]

Table 1: Wyckoff site: `1a`, site symmetry: `-3'm.`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `1b`, site symmetry: `-3'm.`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `3m.`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4,5,6]`
2	$[0,\ 0,\ -z]$	`[7,8,9,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3,4,5,6]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9,10,11,12]`

Table 5: Wyckoff site: `3e`, site symmetry: `.2'/m.`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,7,10]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,5,8,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,6,9,12]`

Table 6: Wyckoff site: `3f`, site symmetry: `.2'/m.`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[1,4,7,10]`
2	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[2,5,8,11]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[3,6,9,12]`

Table 7: Wyckoff site: `6g`, site symmetry: `.2'.`

No.	position	mapping
1	$[x,\ 0,\ 0]$	`[1,7]`
2	$[0,\ x,\ 0]$	`[2,8]`

continued ...

Table 7

No.	position	mapping
3	$[-x, -x, 0]$	`[3,9]`
4	$[-x, 0, 0]$	`[4,10]`
5	$[0, -x, 0]$	`[5,11]`
6	$[x, x, 0]$	`[6,12]`

Table 8: Wyckoff site: `6h`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,7]`
2	$\left[0, x, \frac{1}{2}\right]$	`[2,8]`
3	$\left[-x, -x, \frac{1}{2}\right]$	`[3,9]`
4	$\left[-x, 0, \frac{1}{2}\right]$	`[4,10]`
5	$\left[0, -x, \frac{1}{2}\right]$	`[5,11]`
6	$\left[x, x, \frac{1}{2}\right]$	`[6,12]`

Table 9: Wyckoff site: `6i`, site symmetry: `.m.`

No.	position	mapping
1	$[x, -x, z]$	`[1,5]`
2	$[x, 2x, z]$	`[2,6]`
3	$[-2x, -x, z]$	`[3,4]`
4	$[2x, x, -z]$	`[7,12]`
5	$[-x, x, -z]$	`[8,10]`
6	$[-x, -2x, -z]$	`[9,11]`

Table 10: Wyckoff site: `12j`, site symmetry: `1`

No.	position	mapping
1	$[x, y, z]$	`[1]`
2	$[-y, x-y, z]$	`[2]`
3	$[-x+y, -x, z]$	`[3]`
4	$[-x+y, y, z]$	`[4]`
5	$[-y, -x, z]$	`[5]`
6	$[x, x-y, z]$	`[6]`
7	$[x-y, -y, -z]$	`[7]`
8	$[y, x, -z]$	`[8]`
9	$[-x, -x+y, -z]$	`[9]`
10	$[-x, -y, -z]$	`[10]`
11	$[y, -x+y, -z]$	`[11]`
12	$[x-y, x, -z]$	`[12]`